

Akwasi Poku Ofori Agyei

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EDUCATION

Kwame Nkrumah University of Science and Technology

B. Sc. Computer Science

Kumasi, Ghana

Sep. 2022 – present

Grade: 79.63/100 (First Class) $\approx$ 3.742/4.0 CGPA

EXPERIENCE

Software Engineer Intern

Sep. 2025 – Nov. 2025

Npontu Technologies

Tse Addo, Accra

- Conducted a comparative benchmarking of an in-house language model with SoTA models (GPT-5, Gemini, and LLaMA 3) using prompt based evaluation and qualitative analyses which exposed performance gaps and cultural biases in the African context
- Contributed to building a scalable full-stack web application supporting thousands of concurrent user activity, integrating secure authentication and payment processing while improving system reliability and performance under load.
- Implemented component-based frontend architectures for a mobile application, with structured state management. This optimized the rendering logic, reducing unnecessary re-renders and improving app responsiveness.
- Collaborated with an engineering team, applying structured development practices and version control.

Information Technology Support Intern

Sep. 2024 – Nov. 2024

Gigcorp Limited

Ridge, Accra

- Contributed to data collection and preprocessing for geolocation software, performing data validation and cleaning of raw data to improve data quality
- Collaborated with a team to support and debug the Ghana Police eServices platform, applying root-cause analysis and log driven evaluation to identify system failures and ensure system behaviour aligned with system specifications.

Information Technology Support Intern

Oct. 2023 – Dec. 2023

Gigcorp Limited

Ridge, Accra

- Assisted senior engineers on system support and debugging tasks.
- Contributed to the analysis of software behavior, focusing on system security and scalability.

PROJECTS

SmartCards | *Python, Flask, React, Supabase* [GitHub]

June 2025

- Developed an AI-driven study aid leveraging OpenAI's GPT-3.5-turbo to generate flashcards and quiz questions from uploaded study materials.
- Implemented a data preprocessing pipeline to validate and clean raw input data, extracting the relevant information and removing noise.
- Designed and implemented a full-stack architecture with Flask REST APIs and React frontend, optimizing data flow and system responsiveness for AI-generated content.
- Integrated Supabase for secure user authentication and cloud-based storage, enabling scalable management of user data and study content.
- Explored prompt engineering techniques to improve the accuracy, relevance, and quality of the generated study materials.

EduBridge | *React, Tailwind CSS, Git* [Demo]

May 2023

- Designed and implemented a modular, component-based frontend using React and Tailwind CSS, enabling scalable and maintainable UI development.
- Implemented state management to maintain consistent data flow and improve application efficiency.
- Collaborated in a team environment, applying version control (Git) and agile practices to ensure smooth integration.
- Conducted iterative testing and UI optimization to enhance the usability and overall user experience of the application.

PRESENTATIONS

3rd Year Mini Project Presentation: **SmartCards** [Slides]

3rd Year Web Development Project Presentation: **EduBridge** [Demo]

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks: Keras, React, Node.js, Flask,

Developer Tools: Git, VS Code, Visual Studio, PyCharm, WSL

Libraries: Pandas, NumPy, JAX